

# DBMS PYQ 2022–23 – Full Solutions

Below are **accurate, exam-oriented, point-wise answers** for all questions. Use directly for writing in your answer sheet.

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## Q1 A] Define DBMS. Explain advantages of DBMS over file system. (4 marks)

### Definition of DBMS:

A Database Management System (DBMS) is a software system that allows users to create, store, organize, retrieve and manage data efficiently in a structured way.

### Advantages over File System:

1. **Data Redundancy Control:** DBMS minimizes duplicate data using normalization.
  2. **Data Consistency:** Changes made once are reflected everywhere.
  3. **Data Security:** Provides authentication and access control.
  4. **Concurrency Control:** Many users can access the database simultaneously.
  5. **Backup & Recovery:** Automatic backup and recovery mechanisms.
  6. **Data Integrity:** Ensures accuracy and validity of data.
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## Q1 B] ER model for library system (6 marks)

### Entities:

- **Student(Student\_ID, Name, Fee\_Status)**
- **Book(Book\_ID, Title, Author)**
- **Library(Lib\_ID, Lib\_Name)**

### Relations:

- **Enrollment:** Student enrolls in library (Many-to-One)
- **Borrow:** Student borrows multiple books (M:N)

### Assumptions:

- A student pays a fee → becomes a member.
- A book has a unique ID.
- A student may borrow multiple books.
- A library contains several books.

(ER diagram description — includes Student → Enrollment → Library → Contains → Book → Borrow → Student)

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## Q2 A] Primary key, Foreign key, Candidate key, Super key (4 marks)

### Primary Key:

A unique identifier for a tuple.

- Example: Roll\_No in Student(Roll\_No, Name, Class)

### Foreign Key:

A field in one table that refers to primary key of another.

- Example: Dept\_ID in Employee refers to Department table.

### Candidate Key:

Minimal set of attributes that can uniquely identify tuples.

- Example: {Aadhar\_No}, {Email} both candidate keys.

### Super Key:

A set of attributes that can uniquely identify tuples.

- Example: {Aadhar\_No, Name} is a superkey.
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## Q2 B] Need of normalization + 1NF (6 marks)

### Need of normalization:

1. Remove redundancy
2. Avoid anomalies (Insert, Update, Delete)
3. Improve consistency
4. Better organization of data

### 1NF Definition:

A relation is in **1NF** if it contains **atomic values only** (no multivalued or composite attributes).

### Example:

#### Unnormalized Table:

Student(Name, Phone)

- John, {9999, 8888}

#### 1NF Table:

Student(Name, Phone)

- John, 9999
- John, 8888

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## OR – Q2 B] Check 3NF & convert (6 marks)

Given table: EMP(EMP\_ID, EMP\_NAME, EMP\_ZIP, EMP\_STATE, EMP\_CITY)

#### Functional Dependencies:

- EMP\_ID → EMP\_NAME, EMP\_ZIP
- EMP\_ZIP → EMP\_STATE, EMP\_CITY

#### Violation:

EMP\_ZIP is non-key attribute determining other non-key attributes → **Not in 3NF.**

#### Conversion to 3NF:

1. **EMP Table:** (EMP\_ID, EMP\_NAME, EMP\_ZIP)
2. **ZIP Table:** (EMP\_ZIP, EMP\_STATE, EMP\_CITY)

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## Q3 A] SQL Queries (any 3) (6 marks)

### 1. Names of all branches from loan relation, remove duplicates:

```
SELECT DISTINCT branch_name FROM loan;
```

## 2. Name, loan number, amount of customers having loan at Perryridge:

```
SELECT borrower.cust_name, loan.loan_no, loan.amount
FROM borrower
JOIN loan ON borrower.loan_no = loan.loan_no
WHERE loan.branch_name = 'Perryridge';
```

## 3. Average account balance at each branch:

```
SELECT branch_name, AVG(balance) AS avg_balance
FROM account
GROUP BY branch_name;
```

## 4. All customers with a loan or an account or both:

```
SELECT cust_name FROM depositor
UNION
SELECT cust_name FROM borrower;
```

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## OR – Q3 A] Trigger + example (6 marks)

### Definition:

A trigger is a stored PL/SQL block that is automatically executed when an event occurs on a table.

### Example:

```
CREATE TRIGGER balance_check
BEFORE INSERT ON account
FOR EACH ROW
BEGIN
    IF :NEW.balance < 0 THEN
        RAISE_APPLICATION_ERROR(-20001, 'Balance cannot be negative');
    END IF;
END;
```

### Q3 B] DDL & DML commands with examples (4 marks)

#### DDL (Data Definition Language):

- CREATE: `CREATE TABLE student(id INT, name VARCHAR2(20));`
- ALTER: `ALTER TABLE student ADD age INT;`
- DROP: `DROP TABLE student;`

#### DML (Data Manipulation Language):

- INSERT: `INSERT INTO student VALUES(1, 'Amit');`
  - UPDATE: `UPDATE student SET age=20 WHERE id=1;`
  - DELETE: `DELETE FROM student WHERE id=1;`
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### Q4 A] Transaction states + sequences (6 marks)

#### States:

1. Active
2. Partially Committed
3. Committed
4. Failed
5. Aborted
6. Terminated

#### Valid Sequences:

- Active → Partially Committed → Committed → Terminated
  - Active → Failed → Aborted → Terminated
  - Active → Partially Committed → Failed → Aborted → Terminated
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### Q4 B] ACID Properties (4 marks)

1. **Atomicity:** All or none rule.
  2. **Consistency:** Transaction maintains database rules.
  3. **Isolation:** Transactions execute independently.
  4. **Durability:** After commit, changes persist.
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### Q5 A] Distributed DBMS + types (6 marks)

#### Definition:

A distributed DBMS manages a single logical database stored at multiple physical locations.

## Types:

1. **Homogeneous:**
  2. Same DBMS software at all sites.
  3. **Heterogeneous:**
  4. Different DBMS software.
  5. **Client-server:**
  6. Server stores data, clients request.
  7. **Peer-to-peer:**
  8. All sites equal.
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## Q5 B] Intra-query vs Inter-query parallelism (4 marks)

### Intra-query Parallelism:

Parallel execution **within** a single query.

- Example: parallel scan, sorting.

### Inter-query Parallelism:

Multiple queries executed **in parallel**.

- Example: many users firing queries simultaneously.
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## Q6 A] NoSQL + OLAP (6 marks)

### NoSQL Databases:

- Non-relational databases.
- Types: Key-value, Column, Document, Graph.
- Used for big data, scalability.

### OLAP:

- Online Analytical Processing.
  - Used for analysis, multidimensional queries.
  - Supports slicing, dicing, roll-up.
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## Q6 B] Architecture of Data Warehouse (4 marks)

### Main Components:

1. **Data Sources** (Operational DBs, external files)
  2. **ETL Process** (Extract, Transform, Load)
  3. **Data Warehouse Storage** (central repository)
  4. **Metadata** (data about data)
  5. **OLAP Engine**
  6. **Front-end Tools** (reports, dashboards)
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